

DAY 7

ADS

## Bubble Sort

Initial Array:  $[5, 1, 4, 2, 8]$

Pass 1

- $5 > 1 \rightarrow \text{Swap} \rightarrow [1, 5, 4, 2, 8]$
- $5 > 4 \rightarrow \text{Swap} \rightarrow [1, 4, 5, 2, 8]$
- $8 > 2 \rightarrow \text{Swap} \rightarrow [1, 4, 2, 5, 8]$
- $5 < 8 \rightarrow \text{No Swap}$

After pass 1:  $[1, 4, 2, 5, 8]$

pass 2

- $1 < 4 \rightarrow \text{No Swap}$
- $4 > 2 \rightarrow \text{Swap} \rightarrow [1, 2, 4, 5, 8]$
- $4 < 5 \rightarrow \text{No Swap}$

After pass 2:  $[1, 2, 4, 5, 8]$

Pass 3

- $1 < 2 \rightarrow \text{No Swap}$
- $2 < 4 \rightarrow \text{No Swap}$
- No Swaps  $\rightarrow \text{stop}$

Comparisons = 9  
Swaps = 4

Final Array =  $[1, 2, 4, 5, 8]$

## Selection Sort

Initial Array:  $\begin{matrix} 0 & 1 & 2 & 3 & 4 \\ [64, 25, 12, 22, 11] \end{matrix}$

pass 1: find Smallest  $\rightarrow 11$

Swap  $64 \leftrightarrow 11$

$[11, 25, 12, 22, 64]$

pass 2: find Smallest  $\rightarrow 12$

Swap  $25 \leftrightarrow 12$

$[11, 12, 25, 22, 64]$

pass 3: find Smallest  $\rightarrow 22$

Swap  $25 \leftrightarrow 22$

$[11, 12, 22, 25, 64]$

pass 4: find Smallest  $\rightarrow 25$

Already in position - no Swap

$[11, 12, 22, 25, 64]$

final

$[11, 12, 22, 25, 64]$

Comparisons: 10

Swap: 3

## Insertion Sort

Initial Array:

[12, 11, 13, 5, 6]

$i = 1$ ,  $key = 11$

- $12 > 11 \rightarrow$  Shift 12 Right
- insert 11

[11, 12, 13, 5, 6]

$i = 2$ ,  $key = 13$

- ~~key~~<sup>12</sup>  $\leq 13 \rightarrow$  Stop
- insert 13

[11, 12, 13, 5, 6]

$i = 3$ ,  $key = 5$

- $13 > 5 \rightarrow$  Shift
- $12 > 5 \rightarrow$  Shift
- $11 > 5 \rightarrow$  Shift
- insert 5

[5, 11, 12, 13, 6]

$i = 4$ ,  $key = 6$

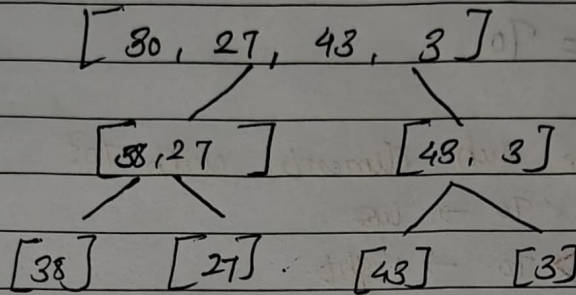
- $13 > 6 \rightarrow$  Shift
- $12 > 6 \rightarrow$  Shift
- $11 > 6 \rightarrow$  Shift
- $5 \leq 6 \rightarrow$  Stop
- insert 6

[5, 6, 11, 12, 13]

# Merge Sort

initial Array:-  $[80, 27, 43, 3]$

## 1. Divide



## 2. Merge

- merge  $[80]$  and  $[27]$ :

$27 < 80 \rightarrow \text{take } 27$

$[27, 80]$

- merge  $[43]$  and  $[3]$ :

$3 < 43 \leftarrow \text{take } 3$

$[3, 43]$

## 3. Final Merge

merge:

-  $3 < 27 \rightarrow 3$

-  $27 < 43 \rightarrow 27$

-  $80 < 43 \rightarrow 80$

- Remaining  $\rightarrow 43$

Comparison = 5

Final  $\rightarrow [3, 27, 80, 43]$

## Quick Sort

Given:  $[10, 80, 30, 90, 40, 50, 70]$

Last element is pivot.

Partition 1

pivot = 70

Compare each element with 70:

- $10 < 70 \rightarrow \text{left}$
- $80 > 70 \rightarrow \text{right}$
- $30 < 70 \rightarrow \text{left}$
- $90 > 70 \rightarrow \text{right}$
- $40 < 70 \rightarrow \text{left}$
- $50 < 70 \rightarrow \text{left}$

After Partition:

$[10, 30, 40, 50, 70, 90, 80]$

Pivot 70 is now in its final position.

- Left part

$[10, 30, 40, 50]$

pivot = 50

All elements are smaller:

$[10, 30, 40, 50]$

pivot 50 is fixed

- Left part again

$[10, 30, 40]$

pivot = 40

$[10, 30, 40]$

pivot 40 is fixed.

Left part

$[10, 80]$

Pivot = 30

$[10, 30]$

Sorted

Right part of 70

$[90, 80]$

Pivot = 80

$[80, 90]$

Sorted

Final Array

$[10, 30, 40, 50, 70, 80, 90]$